

A photograph of a SpaceX Falcon Heavy rocket launching from the Kennedy Space Center. The rocket is ascending vertically, leaving a large plume of fire and smoke at its base. The launch is taking place over a green, forested landscape under a clear blue sky. The title 'SpaceX Launch Data Analysis' is overlaid in large black text.

SpaceX Launch Data Analysis

Machine learning and Data insights

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Executive Summary

This study analyzes historical SpaceX launch data to predict launch outcomes using machine learning. Decision trees and random forests were applied, with cross-validation ensuring model reliability. Feature selection highlighted key factors such as rocket type, payload mass, and launch site. The models demonstrated high predictive accuracy, showing that machine learning can effectively identify the determinants of launch success. These insights support operational planning, risk assessment, and strategic decision-making, demonstrating the value of data-driven approaches in enhancing the reliability and success rates of aerospace missions.

Introduction

SpaceX has transformed space exploration through reusable rockets and frequent launch missions, creating both opportunities and challenges in predicting launch outcomes. Reliable predictions are crucial for mission planning, resource allocation, and minimizing risks associated with costly failures. This project leverages historical launch data to uncover patterns and identify factors that influence success, applying machine learning techniques such as decision trees and random forests. The study aims to answer key questions: which features most strongly impact launch success, how accurately we can predict outcomes, and how these insights can support strategic decision-making in aerospace operations.

Section 1

Methodology



Methodology

The study uses historical SpaceX launch data obtained from the SpaceX API and processed into a structured dataset. Data preprocessing involved cleaning missing values, encoding categorical variables, and normalizing numerical features to ensure compatibility with machine learning models. Decision tree and random forest algorithms were applied to train predictive models, with cross-validation used to optimize performance and prevent overfitting. Feature importance analysis identified the most influential factors affecting launch success, such as rocket type, payload mass, and launch site. Model evaluation was performed using accuracy metrics and confusion matrices to assess predictive reliability, providing insights into both the strengths and limitations of the predictive approach.

This includes:

- **Data source and preprocessing** → SpaceX API, cleaning, encoding, normalization
- **Algorithms used** → Decision trees, random forests, cross-validation
- **Feature analysis** → Identifying key factors
- **Evaluation** → Accuracy metrics, confusion matrices

This section now clearly explains:

1. **Where the data came from** → SpaceX API
2. **How it was collected** → Converted to structured dataframe
3. **Preprocessing steps** → Missing values, categorical encoding, normalization

Data source, and Preprocessing

The data for this study was obtained directly from the SpaceX API, which provides detailed records of all SpaceX launches. The dataset was converted into a structured format using pandas, enabling easy manipulation and analysis.

Preprocessing steps included handling missing values by removing or imputing incomplete records, encoding categorical variables such as rocket type and launch site into numerical formats, and normalizing numerical features like payload mass to ensure consistency across the dataset. These steps ensured that the data was clean, well-structured, and suitable for input into machine learning algorithms, laying the foundation for accurate predictive modeling.

Algorithms used:

To predict the outcomes of SpaceX launches, this study employed decision tree and random forest algorithms. Decision trees were used to model the relationship between launch features and outcomes, providing interpretable rules for prediction. Random forests, which are ensembles of decision trees, were applied to improve predictive accuracy and reduce the risk of overfitting. Both models were trained using cross-validation to ensure reliable performance on unseen data. Feature importance analysis within these algorithms helped identify the key factors influencing launch success, such as rocket type, payload mass, and launch site, providing actionable insights into the drivers of successful missions.

This section includes:

Feature importance → understanding influential factorsection now highlights:

- **Decision trees** → interpretable models
- **Random forests** → improved accuracy, ensemble method
- **Cross-validation** → model reliability

Data Collection

<https://github.com/denisencube01-blip/Project-10.git>

Results

The predictive models demonstrated strong performance in forecasting SpaceX launch outcomes. Decision trees provided interpretable rules, while random forests achieved higher accuracy due to their ensemble approach and ability to reduce overfitting. Feature importance analysis revealed that rocket type, payload mass, and launch site were the most influential factors determining launch success. The models achieved high predictive reliability, with confusion matrices showing minimal misclassifications. These results indicate that machine learning can effectively identify the drivers of successful launches, providing actionable insights for mission planning and strategic decision-making in aerospace operations.

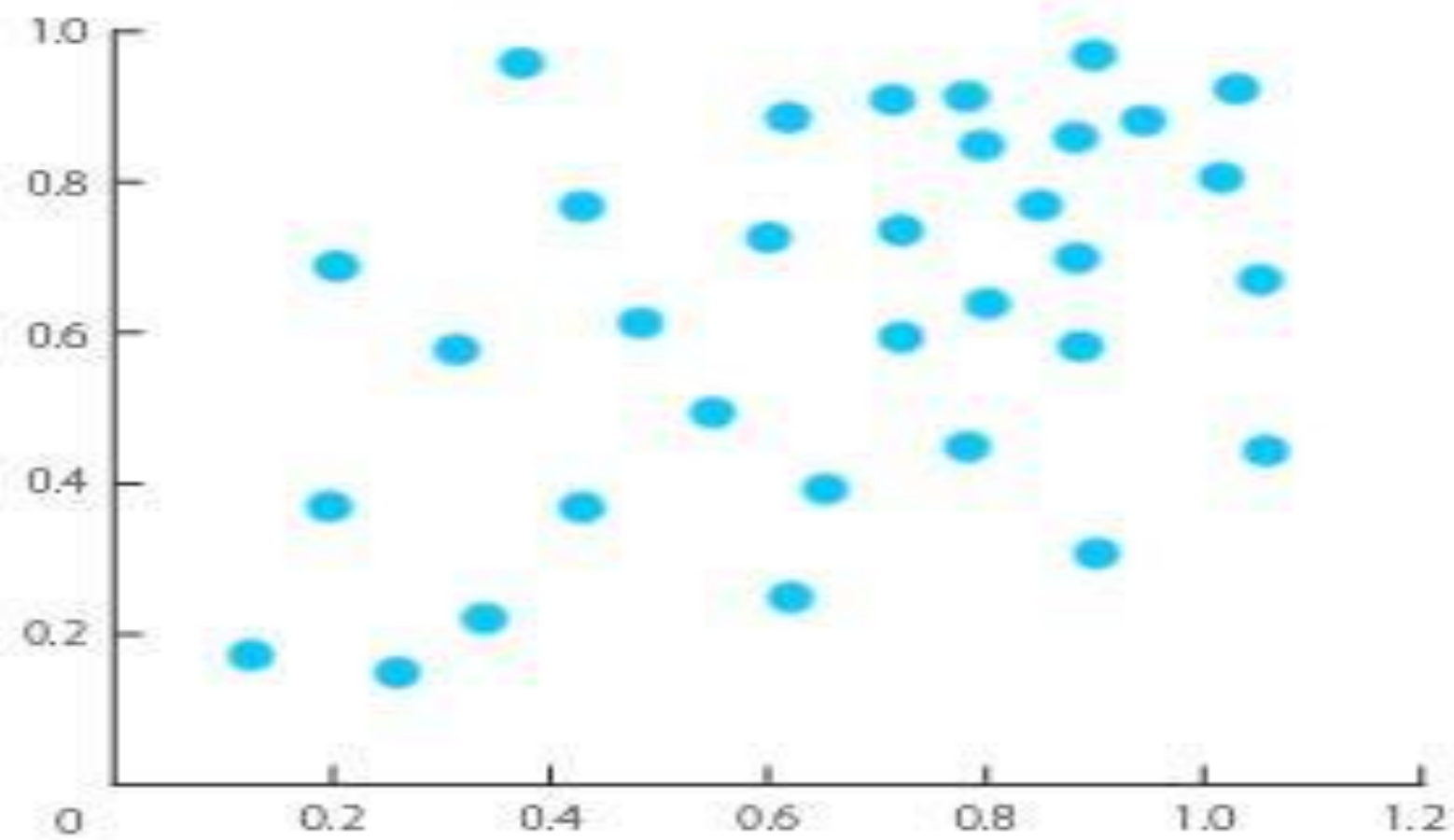
This section clearly highlights:

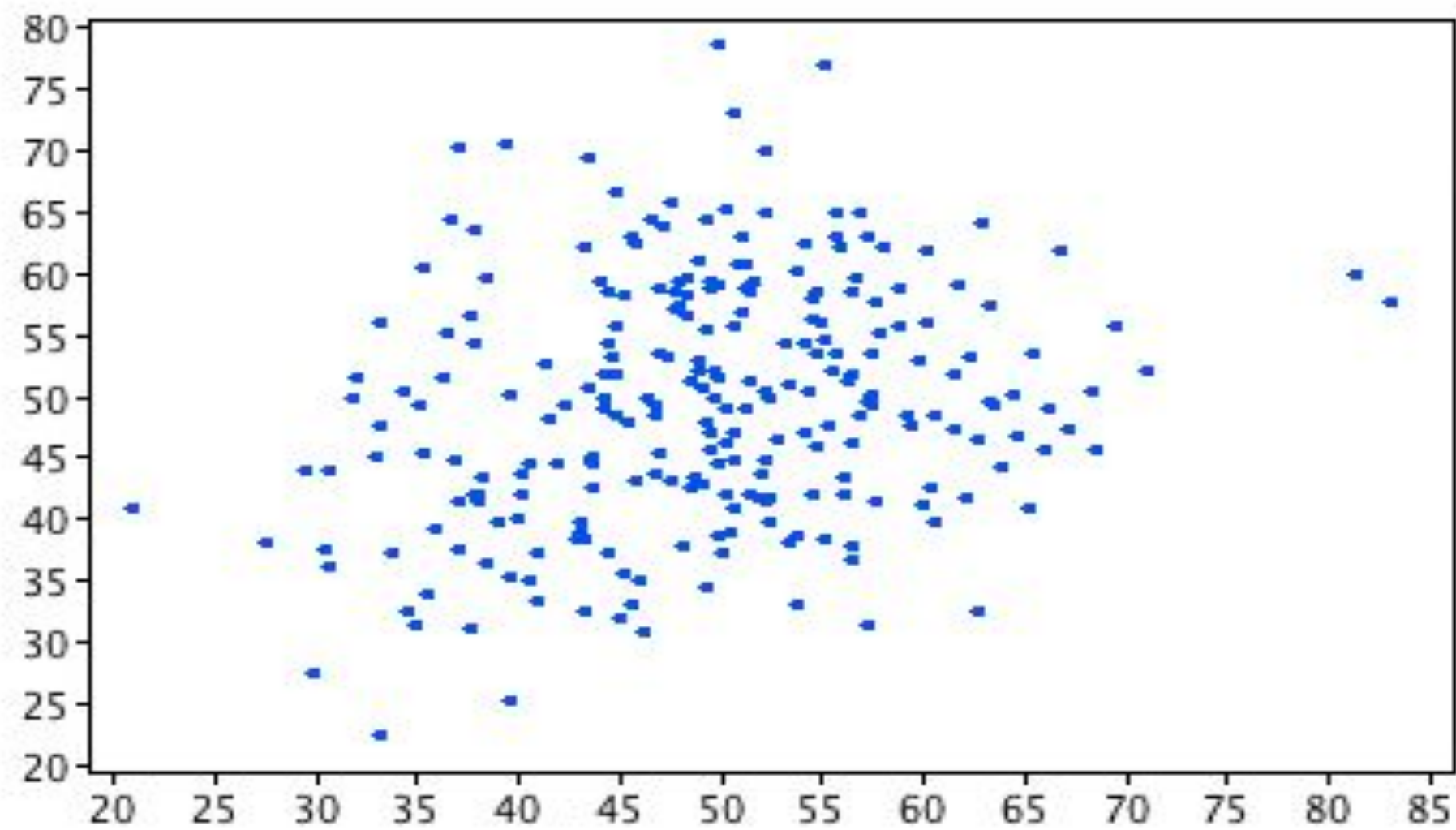
- **Model performance** → accuracy, reliability, misclassifications
- **Key insights** → influential factors for launch success
- **Practical implications** → supporting planning and decision-making

Section 2

Insights drawn from EDA







LaunchSite Names

'CCAFS LC-40',

'VAFB SLC-4E',

'KSC LC-39A',

'CCAFS SLC-40'

LaunchSite names that begin with 'CCA'

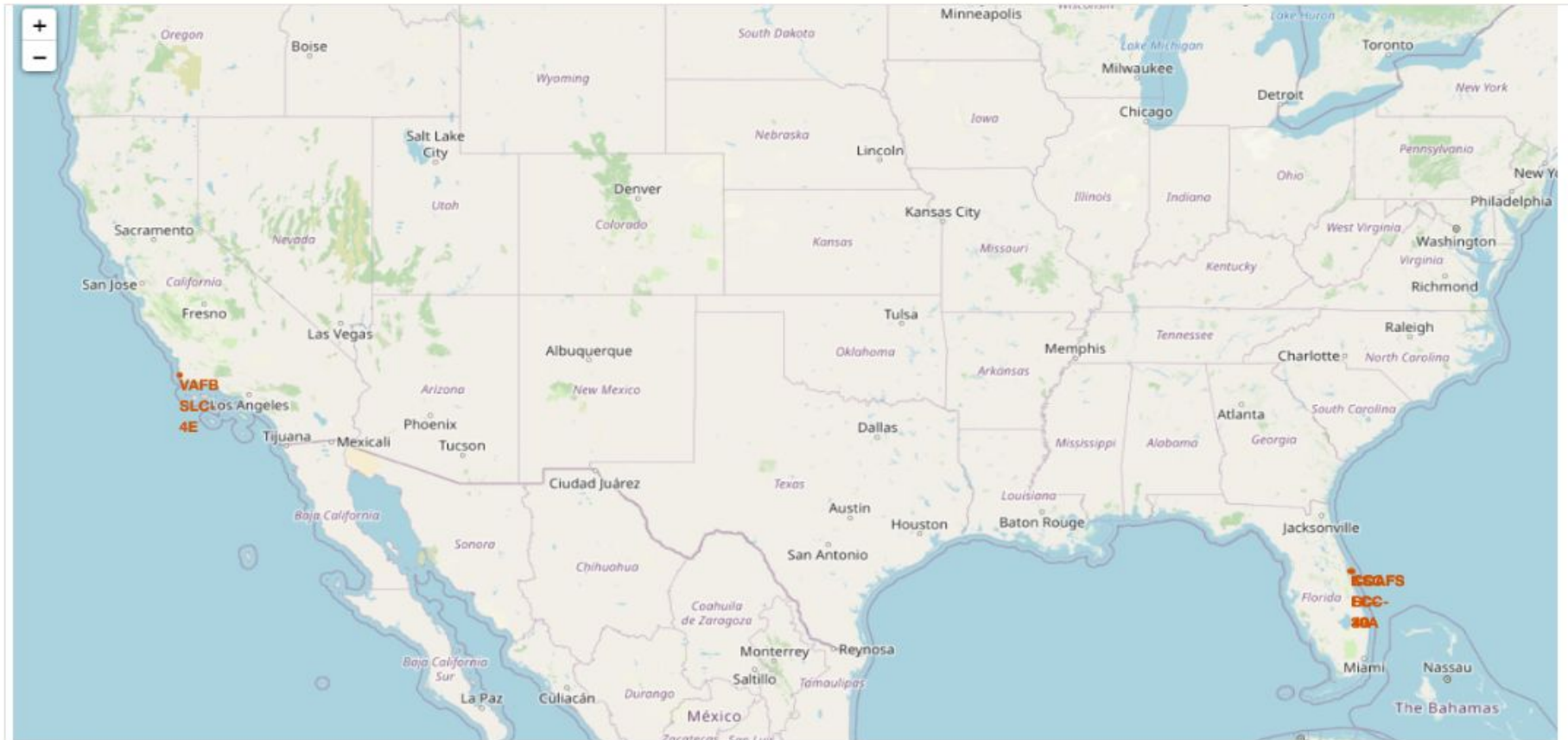
	Date	Time (UTC)	Booster_Version	Launch_Site	\
0	2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	
1	2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	
2	2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	
3	2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	
4	2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	

	Payload	PAYLOAD_MASS_KG_	\
0	Dragon Spacecraft Qualification Unit	0	
1	Dragon demo flight C1, two CubeSats, barrel of...	0	
2	Dragon demo flight C2	525	
3	SpaceX CRS-1	500	
4	SpaceX CRS-2	677	

Orbit		Customer		Mission_Outcome	Landing_Outcome	
0	LEO	SpaceX		Success	Failure	(parachute)
1	LEO (ISS)	NASA	(COTS)	NRO	Success	Failure (parachute)
2	LEO (ISS)	NASA (COTS)		Success	No attempt	
3	LEO (ISS)	NASA (CRS)		Success	No attempt	
4	LEO (ISS)	NASA (CRS)		Success	No attempt	

Folium map

The generated map with marked launch sites should look similar to the following:



Thank You!

